Degala Chenchupradeep

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PROFESSIONAL SUMMARY

Ph.D. Researcher in Quantum Machine Learning with expertise in AI, Deep Learning, and Quantum Computing. Proficient in Python, Qiskit, and PennyLane. Develops quantum-inspired and gate-based models for many-body physics, error mitigation, and QAOA optimization, with applied work in quantum communication (QKD), underwater acoustic networks, quantum-like fuzzy decision making, and clinical QML benchmarking. Ph.D. Scholar at SRM University AP. Publishes reproducible, simulation-based research projects as open-source repositories with pinned dependencies, tests and CI.

RESEARCH INTERESTS

- Quantum machine learning (QML) and variational quantum algorithms: QAOA and VQE.
- Quantum error mitigation and noise-aware algorithm design.
- Tensor networks (matrix product states) for many-body physics.
- Quantum communication and quantum key distribution (QKD).
- Underwater communication.
- Quantum photonics.
- Quantum information theory.

CORE SKILLS

- **Quantum Computing:** Qiskit, PennyLane, Cirq, Tensor Networks (MPS), ZNE (Error Mitigation), VQA, QAOA, VQE.
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Reinforcement Learning, NLP (BERT), GCN.
- **Tools & Languages:** Python, C++, SQL, Django, Flask, Pandas, NetworkX, Big Data (Spark, Hadoop), Cloud Computing.

RESEARCH PROJECTS

Recent open-source research projects (newest first)

- **Chip-aware QAOA placement on modular quantum devices** *GitHub: modular-chip-aware-qaoa-placement | Oct 2026*
  Stack: Python, NumPy, NetworkX, Pauli-trajectory noise simulation, pytest, GitHub Actions.
  – Studied vertex-to-chip assignment for QAOA MaxCut on two-chip modular processors where chip-to-chip coupler gates are 1x, 5x or 20x noisier than intra-chip gates (n=8-12 qubits, 4 graphs per cell, 200 noise trajectories).
  – Compared random placement, Kernighan-Lin bisection and exact minimum bisection with shared noiselessly optimised depth-2 angles, so differences come only from noise.
  – At 20x coupler error, Kernighan-Lin raised the approximation ratio by 0.04-0.06 on average for community and 3-regular graphs (e.g. 0.794 to 0.848 for community graphs at n=8) and cut the mean coupler-gate count from 6.0 to 1.75.
  – Gain was exactly 0 at equal error (control) and 0.009-0.018 at 5x; simulation only, no hardware or quantum-advantage claim.
- **Break-even noise for constrained XY-mixer vs penalty QAOA** *GitHub: xy-mixer-vs-penalty-qaoa-noise-crossover | Oct 2026*
  Stack: Python, NumPy, SciPy (COBYLA), NetworkX, Pauli-trajectory noise simulation, pytest, GitHub Actions.
  – Measured where the extra two-qubit gates of a constrained XY-ring mixer with Dicke start cancel its benefit over penalty QAOA on densest-k-subgraph (n=8, k=4, 5 random graphs), using a CX-count-weighted depolarising noise model.
  – Depth-2 XY-Dicke QAOA fell from a feasible-weighted approximation ratio of 0.758 (noiseless) to 0.390 at two-qubit error p2=5e-3 and 0.295 at 1e-2, against 0.349 noiseless for penalty QAOA, so the crossover lies between p2=5e-3 and 1e-2.
  – XY-Dicke stayed above penalty QAOA at every tested noise level, but the gap at p2=2e-2 (0.207 vs 0.176) is within one standard deviation.
  – Ablation: XY mixer with a uniform start reached only 0.214 at p2=0, so the Dicke initial state drives the advantage; simulation only.

- **Noise-aware co-design of frequency conversion for heterogeneous quantum links** *GitHub: qfc-codesign-heterogeneous-quantum-links | Oct 2026*
  Stack: Python, PyTorch (autograd, Adam), NumPy, SciPy, pandas, pytest, GitHub Actions.
  – Built a differentiable model of a heralded two-node link (930 nm broadband emitter and 780 nm narrowband memory) with PPLN quantum frequency conversion, Raman pump noise, fibre loss and memory dephasing; the objective is the hashing-bound distillable-entanglement rate.
  – Validated the model against a Fock-space Bell-state-measurement simulation (max error 3.3e-16 in fidelity), numerical quadrature of the overlap integral, and an event-level Monte Carlo (max |z| = 2.03).
  – Co-design over 10 randomised hardware instances with 10% calibration error gave 10.9 +/- 4.6 ebit/h at 100 km versus 2.68 +/- 1.06 for the O-band standard (median gain +312%, 10/10 wins, Wilcoxon p = 0.002); the C-band standard gave zero.
  – Ablation showed wavelength choice (a target near 1440-1470 nm) accounts for almost the whole gain; simulation study with representative hardware parameters, not an experimental demonstration.

Ph.D. & core portfolio projects

- **Hardware-Efficient-QPU-Ansatz | VQE for H_2** *GitHub: Hardware-Efficient-QPU-Ansatz*
  – Implemented VQE for the hydrogen molecule with a hardware-efficient ansatz tailored to Rigetti-style QPU topology.
  – Used native gate sets (Ry, Rz) and nearest-neighbor CNOT constraints; converged to −1.114 Hartree.
- **QEM-ZeroNoise-Extrapolation | Quantum Error Mitigation** *GitHub: QEM-ZeroNoise-Extrapolation*
  – Developed ZNE protocols using circuit folding and Richardson extrapolation on a custom density-matrix simulator.
  – Reduced expectation-value error from 31.5% to 5.1% under simulated depolarizing noise models.
- **QML-TensorNet-ManyBody | MPS Simulation** *GitHub: QML-TensorNet-ManyBody*
  – Built a Matrix Product State (MPS) solver from scratch to simulate the 1D Transverse-Field Ising Model.
  – Analyzed the quantum phase transition at $h = 1.0$, matching exact diagonalization for many-body systems.
- **VQA-Quantum-Dynamics | Trotterized Simulation** *GitHub: VQA-Quantum-Dynamics*
  – Translated Heisenberg XY Hamiltonian dynamics into Trotterized gate circuits using PennyLane.
  – Simulated magnon propagation in a 4-qubit spin chain starting from a Néel state.
- **QML-Advantage-Benchmarking | QAOA vs Classical** *GitHub: QML-Advantage-Benchmarking*
  – Benchmarked QAOA against brute-force MaxCut solvers to define the "crossover regime".
  – Quantified approximation ratios and runtime scaling across graph sizes ($N = 4$ to $N = 10$).

PROFESSIONAL EXPERIENCE

Assistant Professor, Computer Science | Raghu Engineering College *Aug 2024 – Nov 2024*

- Taught AI/ML courses; mentored undergraduate projects in quantum and data science; developed new curriculum.

Backend Web Development Intern | PMN Company/Chegg *Jun 2023 – Aug 2023*

- Built Django/Flask services for education platforms, integrated REST APIs, and optimized database queries.

EDUCATION

- **Ph.D. in Computer Science (Pursuing)** | SRM University AP, India *Ongoing*
  – *Research Focus:* Quantum Machine Learning and QAOA Optimization.
- **M.Tech in Data Science (Gold Medalist)** | JNTU-Gurajada *2022 – 2024*
- **B.Tech in Mechanical Engineering** | Siddharth Institute of Engineering *2019 – 2022*

CERTIFICATIONS & HONORS

- **Gold Medalist** in M.Tech Data Science for academic excellence.
- **Quantum Algorithms & Cryptography** | NPTEL (IIT Madras).
- **Deep Learning** | NPTEL.
- **Python for Data Science** | NPTEL.

CONTACT

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